THE CATEGORY OF TOPOLOGICAL GROUPS HAS COLIMITS, COMPUTED ON THE UNDERLYING GROUPS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The representability criterion of the form used to construct colimits through a functor commuting with limits. Invoked inside the counted proof (source lines 6113-6133). Produces the representing topological group.
- The adjoint-exactness statement used to read off commutation with the forgetful functor. Invoked at the end of the counted proof (source lines 6136-6140). Gives the commutation with the forgetful functor to groups.

Source reference. Original reference categories-lemma-a-version-of-brown: sources/stacks-topology/pinned-original-source.tex, source lines 6099–6099. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-a-version-of-brown, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *The category of topological groups has colimits and colimits commute with the forgetful functor to the category of groups.*

Proof. We will use the argument of Categories, Remark categories-remark-how-to-use-it to prove existence of colimits. Namely, suppose that $\mathcal{I} \to \mathit{Top}$, $i \mapsto G_i$ is a functor into the category *TopGroup* of topological groups. Then we can consider

$$F : \mathit{TopGroup} \longrightarrow \mathit{Sets}, \quad H \longmapsto \lim_{\mathcal{I}} \mathrm{Mor}_{\mathit{TopGroup}}(G_i, H)$$

This functor commutes with limits. Moreover, given any topological group H and an element $(\varphi_i : G_i \to H)$ of $F(H)$, there is a subgroup $H' \subset H$ of cardinality at most $|\coprod G_i|$ (coproduct in the category of groups, i.e., the free product on the G_i) such that the morphisms φ_i map into H'. Namely, we can take the induced topology on the subgroup generated by the images of the φ_i. Thus it is clear that the hypotheses of Categories, Lemma categories-lemma-a-version-of-brown are satisfied and we find a topological group G representing the functor F, which precisely means that G is the colimit of the diagram $i \mapsto G_i$.

To see the statement on commutation with the forgetful functor to groups we will use Categories, Lemma categories-lemma-adjoint-exact. Indeed, the forgetful functor has a right adjoint, namely the functor which assigns to a group the corresponding chaotic (or indiscrete) topological group. □